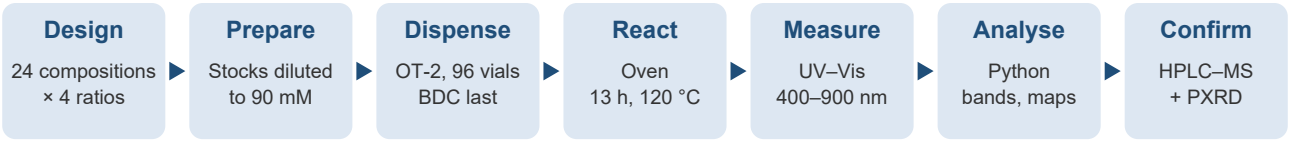
Exploring Heterometallic Co–Ni–Cu Benzenedicarboxylate Coordination Polymers via Automated Ternary Gradient Synthesis

Workflow summary | SDL Hackathon, UZH HTEL, 26–30 October 2026
Team MOFKeKing (project M3-MOF): Aritra Roy, Piyush Ranjan Maharana, Tarak Nath Das



Step	What happens	Who / what
1. Design	Choose 24 Co:Ni:Cu mixes and four metal : ligand ratios; calculate volumes and stock masses.	Python, Excel
2. Prepare	Weigh the metal acetates and BDC, make stocks in DMF and dilute them to 90 mM.	HTEL staff, by hand
3. Dispense	Add DMF, Co, Ni and Cu to 96 GC vials, then BDC last so all reactions start together.	Opentrons OT-2
4. React	Cap the vials and heat for 13 h at 120 °C (oven, as no high-throughput microwave was available).	Oven, by hand
5. Measure	Take a 50 µL sample of each vial and record a UV–Vis spectrum.	Tecan plate reader
6. Analyse	Separate bands from scattering, map them on the composition triangle, test for three-metal features.	Python
7. Confirm	Run HPLC–MS and PXRD on the most promising compositions.	Pending

Status: steps 1–6 are done; step 7 is pending. Code, data and figures are in the M3-MOF repository, and *python analysis/analyse_uvvis.py* reproduces every figure from the raw plate-reader export.